

Business-Oriented Customer Segmentation using Unsupervised Machine Learning for Targeted Marketing

A Data-Driven Approach to Wholesale Customer Analysis

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“Not every customer contributes equally to a business. Understanding who your customers are, what they purchase, and how they behave is the foundation of effective marketing and sustainable business growth.”

1. Executive Summary:

In today’s competitive wholesale market, understanding customer purchasing behavior is essential for improving marketing effectiveness, inventory management, and customer relationship strategies. Treating all customers as a single group often leads to inefficient resource allocation, generalized promotional campaigns, and missed business opportunities.

This project focuses on developing a **Business-Oriented Customer Segmentation Model using Unsupervised Machine Learning techniques**. By analyzing customers based on their annual spending across different product categories—including Fresh, Milk, Grocery, Frozen, Detergents & Paper, and Delicatessen products—the project aims to identify naturally occurring customer segments with similar purchasing behaviors.

The project follows a structured data science workflow consisting of business understanding, exploratory data analysis (EDA), data preprocessing, feature engineering, clustering, cluster evaluation, and business interpretation. Rather than focusing solely on model development, the project emphasizes translating analytical findings into actionable business recommendations.

The identified customer segments can help organizations optimize marketing campaigns, improve inventory planning, personalize customer engagement strategies, strengthen customer retention, and support strategic business decision-making. By combining machine learning techniques with business analytics, this project demonstrates how data-driven customer segmentation can create measurable business value.

2. Objectives:

The primary objective of this project is to develop a business-oriented customer segmentation model using unsupervised machine learning techniques. By grouping customers according to their purchasing behavior, the project aims to help businesses better understand customer needs,

optimize marketing strategies, improve inventory management, enhance customer satisfaction, and support data-driven decision-making.

3.Business Problem Statement:

Many wholesale businesses apply identical marketing strategies to all customers despite significant differences in purchasing behavior. This generalized approach often results in ineffective promotional campaigns, inefficient inventory planning, lower customer engagement, and reduced profitability.

This project addresses the problem by identifying groups of customers with similar purchasing patterns using machine learning techniques. The resulting customer segments enable businesses to design personalized marketing strategies, optimize resource allocation, improve customer relationship management, and make more informed business decisions.

4. Scope of the Project:

The scope of this project includes the analysis of customer purchasing behavior using annual spending data across six product categories. The project covers data exploration, preprocessing, feature scaling, customer segmentation using unsupervised machine learning, cluster evaluation, and business interpretation of the resulting customer groups. The primary focus is on generating actionable business insights rather than merely developing a clustering model.

5. About Dataset:

The dataset used in this project is the **Wholesale Customers Dataset** obtained from the UCI Machine Learning Repository. It contains annual spending data of wholesale customers across six product categories: Fresh, Milk, Grocery, Frozen, Detergents_Paper, and Delicatessen. Each record represents a customer and their purchasing behavior measured in monetary units. The objective is to identify groups of customers with similar purchasing patterns using unsupervised learning techniques.

6. Data Understanding:

- Total observations: **440**
- Features used:
 1. Fresh

2. Milk
 3. Grocery
 4. Frozen
 5. Detergents_Paper
 6. Delicassen
- Channel and Region were excluded from clustering.
 - No missing values were found.

7.Data Cleaning:

- Missing values checked
- Duplicate records checked
- Outliers detected but retained because they represent genuine high-value customers.

8. Exploratory Data Analysis:

Key Findings

- All spending variables showed strong right-skewness.
- Several high-value outliers were observed.
- Milk, Grocery, and Detergents_Paper showed relatively strong positive correlations.
- Fresh spending exhibited weaker correlation with grocery-related products.

9. Data Preprocessing Strategy:

Scaling Methods Considered

Explain

- StandardScaler
- MinMaxScaler
- RobustScaler

Method	Advantages	Disadvantages	Suitable for this dataset?	Reason
StandardScaler	Centers features around zero and scales to unit variance	Sensitive to outliers	Yes	Based on EDA findings
MinMaxScaler	Scales features to a fixed range	Strongly affected by outliers	No	Compare with distribution
RobustScaler	Uses median and IQR, reducing the influence of outliers	Does not normalize the distribution	No	Evaluate using boxplots

Exploratory Data Analysis revealed that all spending variables exhibited significant right-skewness and contained numerous high-value observations. Since K-Means relies on Euclidean distance, feature scaling was necessary to prevent variables with larger magnitudes from dominating the clustering process. Furthermore, because the dataset contained several extreme spending values representing legitimate high-value wholesale customers, these observations were retained rather than removed. Appropriate preprocessing techniques were therefore selected to preserve business information while ensuring effective clustering.

10. Principal Component Analysis (PCA):

Purpose

- Reduce dimensionality.
- Remove redundancy.
- Improve clustering efficiency.
- Enable visualization.

PCA retained four principal components explaining approximately **92% of the total variance**, providing a compact representation of the original dataset while preserving most of the information.

11. Log Transformation:

Objective:

The histogram analysis revealed that all numerical spending variables were highly right-skewed with long tails caused by high-spending wholesale customers. Since K-Means is sensitive to the magnitude of feature values, a logarithmic transformation was applied to reduce skewness while preserving the relative ordering of customers.

Method:

The `numpy.log1p()` function was used, which computes $\log(1 + x)$. This function safely handles zero values while compressing large numerical values more than smaller ones.

12. Clustering Algorithm:

K-Means was used for clustering because of:

1. Euclidean distance
2. Fast
3. Efficient
4. Works well for customer segmentation

13. Choosing Optimal K:

Using the Elbow Method method the elbow occurred at $k=3$

WCSS stands for **Within-Cluster Sum of Squares**. In data science, it is a metric used in **K-Means clustering** to measure how tightly grouped data points are within their respective clusters.

A **lower WCSS** indicates that the data points are very close to their cluster's center (centroid), meaning the clusters are tight and well-defined. Conversely, a higher WCSS means the points are spread out and the clusters are less cohesive

Silhouette Score is a metric used to evaluate the quality and consistency of clusters. It measures how similar a data point is to its own assigned cluster compared to other clusters, producing a score between -1 and +1

The highest practical silhouette score supported selecting three clusters.

The silhouette scores generated:

[0.31582348127480225,
0.2881971308101351,
0.2103992133353044,
0.22208303863312304,
0.22376439587505711,
0.22559978761982324,
0.21598056517777373,
0.22681872056000377,
0.1992220390006898]

14. Cluster Analysis:

Cluster	Characteristics	Business Meaning
Cluster 0	High Milk, Grocery, Detergents	Retail & Grocery Customers
Cluster 1	High Fresh, Frozen	Fresh Food Buyers
Cluster 2	High across all categories	High-Value Bulk Buyers

15. Business Insights:

Cluster 0 – Retail & Grocery Customers

Characteristics

- High spending on Milk, Grocery, and Detergents.
- Lower spending on Fresh and Frozen products.

Business Interpretation

These customers primarily purchase packaged grocery and household products, making them likely retail or convenience-store type buyers.

Marketing Strategy

- Offer discounts on grocery bundles.
- Introduce loyalty programs for repeat purchases.
- Promote household essentials together.

Cluster 1 – Fresh Food Buyers

Characteristics

- High spending on Fresh and Frozen products.
- Lower spending on Grocery and Detergents.

Business Interpretation

These customers prefer fresh food products and may represent restaurants, cafés, or businesses emphasizing fresh ingredients.

Marketing Strategy

- Provide seasonal offers on fresh produce.
- Bundle Fresh and Frozen items.
- Offer early access to premium fresh products.

Cluster 2 – High-Value Bulk Buyers

Characteristics

- High spending across nearly all product categories.
- Highest overall purchasing level.

Business Interpretation

These are the most valuable customers and likely buy in bulk or serve large operations.

Marketing Strategy

- Assign dedicated account managers.
- Offer premium memberships.
- Provide volume discounts and exclusive promotions.

16. Limitations:

- K-Means assumes spherical clusters.
- Cluster labels depend on the chosen number of clusters.
- PCA reduces dimensionality and may lose some information.
- Spending patterns may change over time.
- Results are based only on transaction data and do not include customer demographics.

17. Future Scope:

- Compare K-Means with Hierarchical Clustering or DBSCAN.
- Include customer demographic information.
- Build a recommendation engine for each segment.
- Deploy the model using Streamlit or Flask.
- Update clusters periodically as new customer data arrives.

Conclusion:

This project segmented wholesale customers using K-Means clustering after preprocessing the data with log transformation, standardization, and Principal Component Analysis (PCA). The Elbow Method and Silhouette Score were used to determine the optimal number of clusters.

Three meaningful customer segments were identified based on purchasing behavior.

Visualizations such as heatmaps, bar charts, and PCA scatter plots helped interpret the clusters. These insights can support targeted marketing, customer retention, and inventory management strategies.

References:

- UCI Machine Learning Repository (Wholesale Customers Dataset)
- Scikit-learn Documentation
- Pandas Documentation
- NumPy Documentation
- Matplotlib Documentation
- Seaborn Documentation